

SAFESTRAIN: REIMAGINING THE COMMON HOUSEHOLD STRAINER

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Abstract

This project examines the design and usability of a clip-on pot strainer and presents a redesign intended to make the task of draining boiling water safer, easier, and more intuitive for a wide range of kitchen users. A detailed description of the original product and its user groups establish how the product functions in everyday cooking and what characteristics users bring to this task. Background on how the product is used outlines the challenges commonly encountered when draining boiling water and highlights the ergonomic factors that influence performance and safety. The design section introduces a full circumference strainer with clips, steam guards, and two forms of handle covers, along with the tradeoffs that accompany added safety and adaptability. Considerations of manufacturability discuss the material selection, component construction, and cost implications for producing multiple parts. An evaluation plan is also proposed to compare the redesigned strainer to a traditional version through both performance measures and user experience ratings. Together, these efforts show the value of human factors engineering in designing a product with user needs in mind, leading to a safer and more comfortable cooking experience.

Product and User Description

The product examined in this project is a silicone clip-on pot strainer used to assist with draining boiling water from a pot. It is typically made of flexible, heat-resistant silicone with perforated holes and a small attachment clip that secures part of the pot's edges. It is commonly used when cooking pasta, vegetables, or other foods that require boiling and draining. When in use, the strainer is clipped to one side of the pot, and the user tilts the pot over a sink so that the boiling water passes through the openings.



Figure 1. Initial Pasta Strainer Product

The primary users of this type of product are everyday home cooks. This includes young adults such as college students who often work in small kitchens and prefer simple, compact cooking tools. It also includes middle-aged adults who cook frequently for themselves or their families and appreciate utensils that reduce the number of steps required to prepare a meal. Any age group that is old enough to cook can be a user of this product. However, since it usually involves boiling water and steam, young children are most likely not primary users.

In addition to age groups, several user characteristics are relevant. Some individuals may have reduced grip strength or limited hand mobility, making lighter and easier tools a necessity. Both left-handed and right-handed users should be able to operate the strainer comfortably, since it attaches to the pot in more than one orientation, but there may be instances, depending on the environment, that one hand may be favored. This product is also used by novice cooks, who benefit from tools that require minimal instruction, and by more experienced cooks who value speed and convenience in the kitchen.

Overall, the clip-on pot strainer is a widely used kitchen tool that supports a common cooking task and serves a broad range of users with different abilities and experience levels. Understanding how people interact with this type of cookware, along with the challenges and preferences they bring to the task, provided meaningful insight that guided our user-centered redesign process.

How the Product Is Used by the Customer and the Market Description

The original “improved” clip-on strainer is intended to simplify the common cooking task of draining boiling water, but its real-world application has many ergonomic limitations and issues. Most customers will attach the strainer to one side of the pot, grip the pot with one or both handles, and rotate the pot over a sink to release the water while keeping the food/pasta within the pot. This action requires a prolonged wrist extension, firm grip strength, careful alignment of the strainer, and the ability to do the action while exposed to hot steam. Many users, especially those that have poor grip and wrist strength, reduced dexterity, or smaller hands, tend to struggle to secure the clip tightly, keep the strainer aligned during rotation, or prevent food from slipping out the uncovered sides of the pot. Steam or hot water exiting the open sides could frequently contact the hands and/or face, which creates discomfort and potential burn risk. The asymmetric clip location also puts users into awkward postures depending on whether the user is left or right-hand dominant as well.

Market trends reflect these challenges. Clip-on strainers are popular because of their compact size, low cost, and seemingly improved design. However, customer reviews highlight the same issues: instability when attached, risk of clips slipping under higher loads, inconsistent fit across different pots, accidental spills, and a lack of protection from steam and hot handles. Competing products try to solve these problems either through full-sized metal/plastic strainers which require a separate tool and may require two or even more hands to safely operate, or bulkier silicone lids that sacrifice visibility and control. Even with the demand for safer, more intuitive solutions, few products integrate all the different ergonomic principles to address the user's limitations and physical constraints that occur with the task.

Our hierarchical task analysis (HTA) emphasizes that customers perform the straining task under time pressure, often in a small kitchen environment with limited counter space, while managing hazards like boiling water, slippery/hot pot surfaces, and lack of steam protection. These issues are especially clear by breaking down the draining process into discrete subtasks where we identified where user error or strain was most likely to occur. For example, the HTA (reference image below) highlighted that aligning the strainer with the pot rim, maintaining a stable grip while lifting a heavy pot, and controlling the pour angle were the steps with the highest risk of heat exposure, slippage, loss of food, or misalignment. These subtasks also require simultaneous management of the load, posture, and actual attention. Improving the ergonomics of this task therefore aligns closely with market needs: consumers want tools that reduce cognitive and physical load, prevent accidental burns or spills, and work reliably across a wide range of pot sizes and user capabilities.

- 0. Drain boiling water using SafeStrain
 - 0.1 Prepare Strainer
 - 0.1.1 Retrieve SafeStrain from storage
 - 0.1.2 Inspect clips, steam guards, and overall condition
 - 0.1.3 Verify pot size compatibility with strainer diameter
 - 0.2 Attach Strainer to Pot
 - 0.2.1 Align the strainer slit with the pot rim
 - 0.2.2 Slide the strainer fully onto the pot edge
 - 0.2.3 Engage locking clips to secure the attachment
 - 0.3 Grasp Pot Handles
 - 0.3.1 Position hands on insulated handle covers (single-handle or dual-handle configuration)
 - 0.3.2 Confirm stable, symmetric grip before lifting
 - 0.4 Lift and Tilt Pot
 - 0.4.1 Lift pot from stove using both hands
 - 0.4.2 Move pot carefully toward the sink
 - 0.4.3 Tilt pot at a controlled angle to initiate draining
 - 0.5 Drain Water Through SafeStrain
 - 0.5.1 Maintain pot alignment to prevent pasta escape
 - 0.5.2 Monitor steam direction and avoid exposure (assisted by steam guards)
 - 0.5.3 Continue tilt until water flow reduces to minimal drips

Figure 2. Hierarchal Task Analysis

Our redesigned SafeStrain addresses these concerns by replacing the partial clip-on design with a full covering strainer around the entire circumference of a pot that locks securely. This should eliminate the most common failure modes of rotational slipping, heat uncomfortability, and tough coordination. Steam guards and optional handle covers directly target the heat hazards that the users face. The stretchable material allows the design to fit multiple different pot sizes, so the user doesn't have to worry about compatibility. Finally, with the design being able to stretch and set itself over the pot it frees the user to use both hands to execute the task effectively, making it easier and more stable. By reducing the strength requirements, improving control during rotation, and minimizing exposure to steam, the SafeStrain enhances both the safety and ease of the task, meeting user needs in the existing marketplace and positioning the design as a more reliable, user-centered alternative to traditional strainers.

Design

The final design of our prototype can be seen in the drawing below:

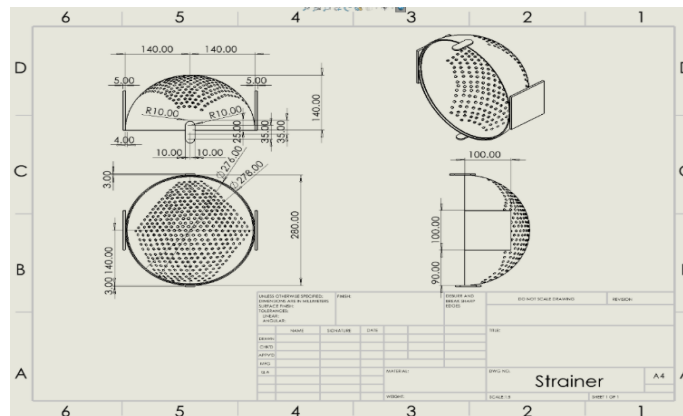


Figure 3. Drawing of Strainer

The design is circular with a radius of 140 mm. This is the typical size of a large pot, so our design will be able to conform to most users' needs. As shown in the drawing, our product includes a variety of features that were outlined in our product description. The strainer will cover the entirety of the pot, making it easier to use, as users will not have to worry about spilling pasta off the side of the strainer. The perimeter of the strainer includes a slit, which would allow users to slide the strainer over the pot. However, our design also includes clips, which will act as a safeguard to prevent the strainer from coming off the pot so that users do not injure themselves.

Additionally, our design includes steam guards which are placed on the sides of the pot where a user's hands would be. This will help to prevent burns that could be caused by steam pouring out the sides when the pasta is being strained. In addition to the strainer itself, our product will include optional handle covers that will help reduce the number of injuries that occur while using a pasta strainer.

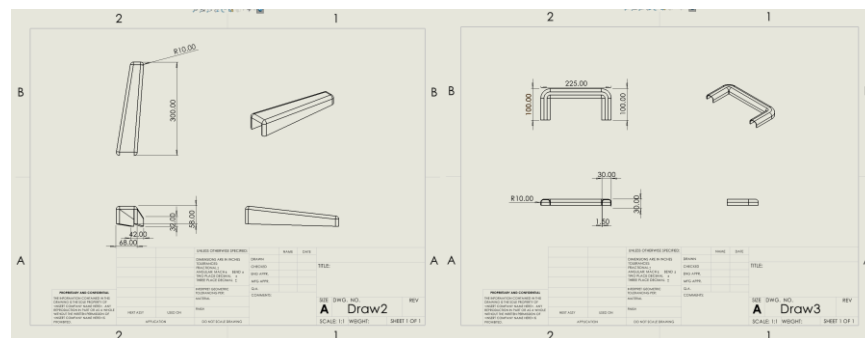


Figure 4. Handle Cover Drawings

There are two types of handle covers: one for a pot with one long handle, and one for a pot with two short handles. These will help guard the user's hands from touching a hot handle, helping to prevent burns.

All of the components of our product will be made of polypropylene. While our design is safer, there are definitely some tradeoffs that exist compared to the original design. It will definitely be more costly to manufacture both types of handle covers. However, it will be more flexible for users to be able to use the product with a variety of different pot types. In addition, there is a space consideration with this design. This design will take up a lot more space in a person's kitchen compared to the original strainer this design is based off of. However, our design includes more features that are meant to keep the user safe and use our design more intuitively.

Manufacturability

Our primary material for the strainer is polypropylene (PP), which is a lightweight, durable thermoplastic commonly used in everyday products. It is especially popular for food utensils because it resists heat, moisture, and many chemicals, making it safe for direct contact with food. PP doesn't easily absorb oils or water, can withstand repeated washing, and is approved for food safety standards. PP has a melting point of about 160-170 degrees Celsius, which means it would be able to withstand very hot food temperatures. Furthermore, it is very easy to customize, hence it would be very easy for manufacturers to easily change colors and make the strainer more aesthetic for customers.

Environmental considerations further support the use of polypropylene because, compared to other materials that could have been selected, PP offers the best balance of durability, recyclability, and low environmental impact for a consumer kitchen product. We also considered other commonly used kitchenware materials. Full silicone construction, for example, provides excellent heat resistance but is significantly harder to recycle, often ends up in landfills, and has a higher per-unit material cost. Stainless steel is highly durable and fully recyclable, but it comes with a much higher carbon footprint during production and adds unnecessary weight and cost for a design that does not require metal strength.

In terms of production, our main technique is Injection molding. Injection molding allows the entire strainer to be produced as a mostly single-piece component, reducing assembly labor and increasing consistency across batches. Proper draft angles, uniform wall thickness, rib support, and thoughtful gate placement help limit warping and shorten cycle time. Furthermore, because the holes in the strainer are small and circular, no additional design stencil would be needed, which also further speeds up the manufacturing process. Another manufacturability advantage of the design is that secondary components, such as handle covers, do not require complex assembly during production. Since these grips are produced as simple slip-on TPE (Thermoplastic Elastomer) sleeves or molded as independent pieces, they can be packaged alongside the main strainer body without needing to be permanently attached on the factory

floor. This greatly reduces assembly labor, shortens production time, and avoids the need for adhesives, screws, or multi-step fastening processes.

To validate performance, manufacturers would use straightforward, non-time-consuming testing techniques that are standard in consumer kitchenware production. Thermal cycling tests involve repeatedly exposing the strainer to boiling water and then cooling it, which quickly confirms that the PP body and clips maintain their shape without warping. Dishwasher-resistance tests are equally simple: the product is run through multiple standard wash cycles to check for discoloration, cracking, or loss of flexibility. Clip fatigue tests use automated open-and-release motions to simulate months of real use within just a few hours, ensuring the living-hinge design remains secure over time. Leakage and flow tests are done by draining water and cooked pasta through the strainer and measuring how much, if any, food escapes, which only takes minutes. Tensile checks around the perforations to verify that thin sections do not tear under normal force. All these procedures rely on basic equipment, require minimal labor, and fit naturally into routine quality-control workflows, making validation efficient even at high production volumes.

Shipping and distribution planning also play an important role in manufacturability. The strainer's lightweight PP construction reduces freight cost, and by designing the product so multiple units can nest together, manufacturers minimize carton size, pallet volume, and emissions across the supply chain. For mass production, suitable manufacturing partners include companies with deep experience in injection-molded consumer kitchenware, such as Plastikon Industries, Berry Global, Wilton Industries, and Precise Plastics. Large contract manufacturers like Mold-Rite Plastics, Custom-Pak, and Flex have the capability to support mold construction, multi-cavity tooling, high-volume molding, quality assurance testing, and packaging integration. These companies also commonly handle regulatory documentation and pilot builds, making them strong candidates for scaling a product from concept to retail distribution.

Lastly, we examined the cost of production. Cost plays a critical role in manufacturing, particularly for consumer products that must deliver good performance while remaining affordable. Factors such as the materials used, the complexity of tooling, the time required for assembly, and the overall shipping weight all contribute to the final expense. By choosing polypropylene and opting for a straightforward, largely single-piece design, we managed to keep production cost-effective. The table below outlines an estimated distribution of the main expenses associated with producing the upgraded pasta strainer at medium to high production volumes.

Table 1. Manufacturability Cost Estimation

Cost Category	Estimated Cost (per unit)
Raw Material (PP Resin)	\$0.15 – \$0.30
Optional TPE Handle Covers	\$0.10 – \$0.25
Injection Molding Process	\$0.20 – \$0.50
Tooling Amortization	\$0.05 – \$0.15
Packaging & Labeling	\$0.10 – \$0.25
Quality Testing & Inspection	\$0.03 – \$0.08
Estimated Cost for 1 Strainer	\$0.68 - \$1.73

In designing our pasta strainer, we carefully considered multiple factors that influence manufacturability, safety, and overall practicality. Production efficiency, environmental impact, and user safety were prioritized, ensuring that our product is not only customer-friendly but also manufacturing- and environmentally friendly, making it industrially feasible.

Evaluation Design

To evaluate the effectiveness of our redesign, we designed a study that compares the redesigned strainer to a traditional model under controlled, repeatable conditions. We will recruit 20–30 adults (ages 18–65) representing typical home-kitchen users. Participants will vary in cooking experience, hand size, and grip strength, with at least 25% of novice users. Individuals with current wrist injuries will be excluded.

The study will manipulate two independent variables: Strainer type (Traditional strainer vs. Our redesigned ergonomic strainer) and pot type (Single-handled vs. Double-handled). The dependent variables include both objective and subjective outcomes. Objective measures will consist of task completion time (seconds), amount of pasta spilled (grams), number of errors (mis-clipping, incorrect alignment, unsafe handling), wrist deviation angle (degree), and perceived exertion (Borg RPE scale). Subjective measures include a usability rating, perceived safety, comfort, and ease of grip.

Participants will perform the following tasks using each strainer (counterbalanced order) and will complete each condition twice to measure learning effects: 1. Clip the strainer onto a pot filled with pasta and hot water (simulated 60–70°C for safety). 2. Attach the handle cover. 3. Carry the pot to the sink. 4. Drain the water fully through the strainer. 5. Remove the strainer safely after cooling.

We expect the redesigned strainer to: Reduce errors and pasta spillage, lower wrist strain and exertion, improve task speed, and increase comfort, safety, and usability scores. Benefits should be especially noticeable for users with smaller hands or less experience.

To analyze data, we will conduct statistical tests, like paired t-tests or repeated-measures ANOVA, to compare traditional vs. our strainer across key DVs; Chi-square tests for differences in error frequency; and subjective ratings will be summarized using descriptive statistics. If the observed results match our predictions, we conclude that redesign improves safety, usability, and performance and should replace current models on the market.